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Graded Assignment • 2h

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Module 2 Introduction

Lesson 1: Estimating Probabilities and Posterior Distributions

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Lesson 3: Normal Distribution and Other Single-Parameter Models

Module 2 Summative Assessment

✓ **Graded Assignment:** Module 2 Summative Assessment
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Module 2 Summary

Your grade: 93.33%

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Module 2 Summative Assessment

Review Learning Objectives

Instructions

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Assignment details

Due
Oct 27, 11:59 PM IST

Attempts
0 left (1 attempts every 8 hours)

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Submitted
Nov 21, 2:01 AM IST

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1. A marketing company is testing a new online advertisement campaign. They start with a prior belief that the proportion of viewers who will click on the ad (engagement) follows a Beta distribution with parameters $\alpha=22$ and $\beta=36$. After running the ad 1000 times, they find that 35% of viewers clicked on the ad. How has their mean engagement changed with this new information? **1 / 1 point**

○ There is not enough information to determine this
● It decreased by 3 percentage points
○ It increased by 3 percentage points
○ It did not change

Correct
Correct! The data provided has a mean of 35%. The prior distribution had a mean of 38%. Because the dataset is so large, it dominates the posterior distribution.

2. Suppose $y|\theta$ is exponentially distributed with rate θ, and the prior distribution of θ is Gamma(α,β). Suppose we know our observations, y, is larger than 200 but we do not know the exact amount. What is the posterior distribution? (Hint: Consider all possible amounts of y larger than 200.) **1 / 1 point**

○ Gamma(α,β)
○ It is not possible to compute.
○ Gamma($\alpha+1,\beta+200$)
● Gamma($\alpha,\beta+200$)

Correct
Correct!

3. Consider two coins, C_1 and C_2, where $p(H|C_1) = 0.75$ and $p(H|C_2) = 0.25$. Choose one coin at random and imagine flipping it twice. Both are tails. What is the expected number of flips (approximately) needed to observe heads? (Hint: What is the probability of each coin?) **1 / 1 point**

● 3.7
○ 1.6
○ 2.5
○ 2.7

Correct
Correct! The posterior probability of C_1 and C_2 are 0.1 and 0.9, respectively. The expected number of flips for each coin is 1.33 and 4, respectively.

4. Suppose your prior distribution for θ, the proportion of Illinois residents who support legal access to abortion, is Gamma with mean 0.7 and standard deviation 0.15. What are the approximate values for α and β? **1 / 1 point**

○ $\alpha=0.214, \beta=0.150$
○ $\alpha=3.27, \beta=4.67$
● $\alpha=21.8, \beta=31.1$
○ There are infinitely many possible values

Correct
Correct! Using the identities of the Gamma for mean and variance, this is what you should get when solving for the two parameters.

5. Suppose you are working with a problem whose likelihood follows a normal with unknown mean μ and known variance σ^2. Furthermore, suppose you use an noninformative prior distribution (defined finitely on the number line). What kind of distribution is the posterior distribution? **1 / 1 point**

○ Exponential
○ Gamma
○ Beta
● None of the above

Correct
Correct! Because the uniform distribution is defined finitely on the number line, it is not conjugate. Multiplied by the normal distribution, we end up with something truncated, which is different entirely.

6. A factory manufactures circuits with a target length of 20 cm. Due to variations in the manufacturing process, there are slight deviations from this target. Historically, the variance of the lengths produced is $\sigma^2=1$ sq. cm. The prior belief for the average length of these circuits is $\mu \sim \mathcal{N}(20.0,1.0)$. **1 / 1 point**

You collect the following data from a sample of rods: $y=\{20.2,19.8,20.5,20.1,20.3\}$.

Based on this sample data, how would you update your prior belief about the mean length of the circuits and determine the posterior distribution of the mean length?

● $\mu|y \sim N(20.15,0.17)$
○ $\mu|y \sim N(20.2,0.7)$
○ $\mu|y \sim N(20.1,0.5)$
○ There is not enough information to compute this.

Correct
Correct!

7. Suppose you are tracking births in an isolated community and out of the 100 births you observe, 49 are female. What is the expected value of θ? **1 / 1 point**

○ .99
● .49
○ Unable to answer the question
○ .48

Correct
Assuming a uniform distribution for θ, we know the posterior distribution is the Beta distribution based on the supplied data. What is this distributions expected value?

8. Suppose you have a Beta(2, 2) prior distribution on the probability θ that a coin will yield a head when flipped. The coin is independently flipped ten times, and heads appear fewer than 3 times. You are not told how many heads were seen, only that the number is less than 3. What is your exact posterior density (up to a proportionality constant) for θ? **1 / 1 point**

○ Beta(3, 10)
○ Beta(2, 10)
● A distribution distinct from a Beta distribution
○ Beta(2, 11)

Correct
The result yields a posterior distribution of the form $\theta\left(1-\theta\right)^9\left[1+8\theta+36\theta^2\right]$

9. Consider a scenario where you are studying the rate of failure of a particular machine component. The time until failure of these components follows an exponential distribution with an unknown rate parameter λ. You have a prior belief about the rate parameter λ, which follows a Gamma distribution with shape parameter $\alpha_0=3$ and rate parameter $\beta_0=2$. You collect a sample of 5 failure times (in hours) for these components: $y=\{2,3,1,4,5\}$. **1 / 1 point**

Based on this data, perform Bayesian inference to determine the posterior distribution of the rate parameter λ.

○ Gamma(3,19)
○ It is not possible to compute with the given information
● Gamma(8,17)
○ Gamma(8,37)

Correct
Correct! We can compute the posterior distribution by considering $a_n = a_o + n$ and $\beta_n = \beta_o + ny$

10. What is the key characteristic of a noninformative prior? **1 / 1 point**

○ It reflects historical data.
● It assumes all values of the parameter are equally likely.
○ It is influenced by prior experiments.

○ It incorporates expert knowledge.

Correct
Correct! Noninformative priors (sometimes called “uninformative”) incorporate no information and therefore see all possible outcomes as equally likely.

11. Suppose you are studying the precision of the heights of a specific plant species. You have a prior belief about the precision, and you collect some sample data to update this belief. The heights of the plants are normally distributed with a known mean $\mu=150$cm but an unknown variance σ^2. You assume a Gamma prior for the precision $\tau=1/\sigma^2$ with shape parameter α_0=2 and rate parameter β_0=1. You collect a sample of 10 plant heights with the following measurements (in cm): $y=\{149,151,152,148,150,153,149,147,151,150\}$. Based on this data, perform Bayesian inference to determine the posterior distribution of the precision τ. (Hint: If the prior is a Gamma distribution, what is the posterior distribution?) **1 / 1 point**

○ Gamma(12, 31)
● Gamma(7, 16)
○ It is not possible to compute with the given information
○ Normal(150, 1)

Correct
Correct! If we treat the prior as a Gamma distribution, the posterior has parameters $\alpha_n = \alpha_0 + \frac{n}{2} and \beta_n = \beta_0 + \frac{s}{2}$

12. What happens to the posterior distribution as the sample size increases? **0 / 1 point**

○ It becomes more influenced by the data.
○ It remains equally influenced by both prior and data regardless of sample size.
● It becomes less variable than the prior distribution but equally variable as the sample size increases.
○ It becomes more influenced by the prior distribution.

Incorrect
The posterior distribution becomes less variable than the prior, but it is not equally variable as the sample size increases; it is more influenced by the data.

13. Which of the following statements best describes an informative prior? **1 / 1 point**

○ It assumes all parameter values are equally likely.
○ It cannot influence the posterior distribution.
○ It is always derived from uniform distribution.
● It is a valid probability distribution function informed by prior knowledge.

Correct
Correct! An informative prior reflects some kind of information held, whether that is based on expert hypotheses or prior experimentation.

14. Suppose $y|\theta$ is exponentially distributed with rate θ, and the prior distribution of θ is Gamma(α,β). Suppose we know our observations, y, is larger than 200 but we do not know the exact amount. What is the posterior distribution? (Hint: Consider all possible amounts of y larger than 200.) **1 / 1 point**

○ Gamma(α,β)
● Gamma($\alpha,\beta+200$)
○ It is not possible to compute.
○ Gamma($\alpha+1,\beta+200$)

Correct
Correct!

15. What is the primary difference between conjugate and nonconjugate priors? **1 / 1 point**

○ Nonconjugate priors are always preferred in practical applications.
● Conjugate priors lead to posterior distributions in the same family as the prior distribution.
○ Nonconjugate priors require less computational effort.
○ Conjugate priors are always more accurate.

Correct
Correct! By ensuring our distributions come from the same family, we’re able to derive the posterior distribution more easily.